MATH 101 Unit 1: Expressions, Equations, and Inequalities - Practice 2

Version 202610122020.35 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Simplify. In (a), x and y are nonzero; give positive exponents.

(a) $\dfrac{6x^{-3}y^{3}}{3xy^{-1}}$

(b) $\sqrt{48}$

2. Simplify and state all values excluded by the original expression.

$$\frac{x^{2}-9}{x^{2}+2x-15}$$

3. (a) Expand and simplify. (b) Factor completely.

(a) $(3x-3)(x+3)$

(b) $2x^{2}-6x-8$

4. Solve both equations. In (b), state original restrictions and check your answer.

(a) $x^{2}+2x-8=0$

(b) $\frac{5}{2x-6}+\frac{1}{x-3}=\frac{7}{2}$

5. Solve the inequality and write the solution in interval notation.

$|2x-1|\le 4$

Student answer sheet

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Question 1

(a) 10 pts

(b) 10 pts

Question 2

Expression 14 pts

Excluded values 6 pts

Student answer sheet

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Question 3

(a) 10 pts

(b) 10 pts

Question 4

(a) 10 pts

(b), with check 10 pts

Question 5

Derived bounds 12 pts

Interval 8 pts

Answer key

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Question 1

(a)	$\frac{2y^4}{x^4}$	10 pts
(b)	$4\sqrt{3}$	10 pts

Question 2

Expression	$\frac{x+3}{x+5}$	14 pts
Excluded values	$\{-5, 3\}$	6 pts

Answer key

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Question 3

(a)	$3x^2 + 6x - 9$	10 pts
(b)	$2(x - 4)(x + 1)$	10 pts

Question 4

(a)	$x = -4,\ 2$	10 pts
(b), with check	$x = 4;\ x \neq 3$	10 pts

Question 5

Derived bounds	$-\frac{3}{2} \leq x \leq \frac{5}{2}$	12 pts
Interval	$[-\frac{3}{2}, \frac{5}{2}]$	8 pts